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Tata Power and India's Energy Transition

In late 2024, Dr. Praveer Sinha, CEO of Tata Power, India's oldest and largest private power player by installed capacity, considered the way forward. He envisioned Tata Power at the forefront of India's energy transition and progress towards net zero by 2070, which hinged on the decarbonization of its highly carbon-intensive power sector, the single largest sectoral contributor to the country's greenhouse gas (GHG) emissions. In 2020, Tata Power boldly announced a commitment to net-zero emissions by 2050, concurrent with a complete phase-down of thermal capacity. It later brought this commitment forward to 2045. To prepare for declining revenues from thermal power generation, it was actively expanding its renewable business, which included power generation and component manufacturing, its new-age energy solutions business, decentralized renewable energy solutions, products, and services to enhance energy efficiency, sustainable mobility infrastructure, and its transmission and distribution business.

Still, stakeholders had concerns about Tata Power's ambitions. Was the firm's decision to sacrifice potentially high returns in thermal power generation financially imprudent, or did it position the firm well as India inevitably accelerated its energy transition? Was its presence across most segments of the power value chain a source of strength or confusion, and as such, should it be more disciplined about focus vs. breadth? While Sinha was a self-professed proponent of technological experimentation, proudly stating that "at any given time, [the firm] had half a dozen pilots that may or may not work out," how should it think about capital allocation in an uncertain technological and returns landscape?

The Global Energy Transition

Power generation from burning fossil fuels constituted the single largest source of global GHG emissions. In advanced economies, power demand was expected to emanate from the ongoing electrification of economies and rising power needs from sectors such as artificial intelligence and electric vehicles. In contrast, in emerging economies, it was likely to come from an intensification of industrialization, rising per-capita incomes, and increased power coverage. Without action, emissions were only set to rise. (See **Exhibit 1** for an overview of the different components of the power industry).

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At the heart of the global power transition was coal, the most abundant fossil fuel with a long history. Used for heating since caveman times, its usage increased during the 1700s Industrial Revolution. Just as its use transformed the global economy with explosive growth in infrastructure and industrial activity, the energy transition away from coal promised to systemically alter how humans lived, commuted, produced, and consumed.

Reducing fossil fuel's share in energy consumption from ~80% to 16% by 2050 while increasing the share of renewables from ~16% to ~77% globally was integral to achieving the Paris Agreement.¹ While the world was spending record amounts on renewables, annual investment

(including on grids and battery storage) was still only half of what it needed to be, while annual spending on energy efficiency and electrification needed to triple for the world to reach net-zero by 2050. Emerging markets, in particular, were spending far less than needed.² In addition, the International Energy Agency noted that despite efforts to shift towards renewable energy, countries still invested in fossil fuels to meet ongoing energy demands.³

India's Net-Zero Goals and Power Sector Transition

Emerging economies faced a unique challenge in ensuring a "just" energy transition as they sought prosperity through increased energy consumption, similar to advanced economies that historically contributed to emissions. This dilemma of achieving "development while decarbonizing" required transitioning traditional energy sector workers to new solutions without compromising their livelihoods and incomes. Perhaps nowhere was this challenge more evident than in India.

In 2024, India was the world's fifth-largest and fastest-growing large economy, on track to become the third-largest by 2030, behind the US and China. Given its high vulnerability to climate change, India was taking big steps to decouple its growth from emissions. Although it ranked lowest amongst G20 countries on power consumption and emissions per capita, in absolute terms, it ranked third highest on both. It was also home to many of the world's most polluted cities.

India's net zero by 2070 commitment hinged on decarbonizing its power sector, which contributed 34% of the country's total emissions. India's grid was highly carbon-intensive, at 725g CO₂/kWh versus a global average of 510, given its traditional reliance on coal-based production. From 2000 to 2021, India's power demand quadrupled, and 900 million citizens gained access to power as the country aspired to roll out universal access.^{4,5} Demand was forecast to quadruple further by 2050.⁶ This increased demand would only further pressure India's grid, so moving to clean energy was key. (See **Exhibit 2** for sector-wise CO₂ emissions in India in 2019.)

To achieve net zero by 2070, India needed an estimated \$3.8 trillion of investments in renewable energy and storage facilities by 2050.⁷ With 14 of the world's 20 most polluted cities in India, the urgency for clean energy action became critical as air pollution increasingly impacted the health and well-being of its population.⁸ In addition, summer temperatures were reaching as high as 50 – 55 degrees Celsius in some regions, creating serious risks to both health and infrastructure.⁹ Despite numerous efforts, India's annual spending on the energy transition was low.¹⁰ It originally projected \$2.5 trillion between 2015 and 2030, but the estimated investment rose sharply; it now needed \$10.1 trillion to make this transition possible. India's complex economic backdrop, with urgent and pressing

^a In 2024, industry accounted for 41% of power demand, the residential sector 25%, agriculture 17%, the commercial sector 8%.

needs for investments in education, health, and infrastructure, did not allow for adequate public financing of its energy transition.

Renewables and Thermal to Coexist

India's target was a 50% share of renewables in power capacity by 2030, with a total non-fossil-fuel capacity of 500GW and energy independence (the ability to meet the nation's energy requirements domestically) by 2047.¹¹ (See **Exhibit 3** for current and forecasted power capacity across sources)

In 2023, renewable energy constituted ~11.5% of power generation and coal was 75.8%.¹² While this was partly reflective of the lower capacity factor – the ratio of energy generated divided by installed capacity – at renewable plants, it was also symptomatic of underlying inefficiencies, including the lack of adequate storage and transmission infrastructure and inflexible power markets.

India had made good strides towards these goals. As of 2024, it had the fourth-highest installed renewable energy generation capacity globally. Renewables accounted for ~32% of the country's power capacity.¹³ In Fiscal Year 2024 (April 2023 – March 2024), more than 70% of power capacity addition came from RE, with solar accounting for 81% of this; nuclear capacity expanded for the first time since FY17, while coal's share fell below 50% for the first time.¹⁴ The government planned to tender 50GW of RE capacity each year to meet its targets. Still, India needed to find ways to integrate its renewable energy better. (See **Exhibit 4** for installed power capacity and power generation by sources)

It was evident that coal would remain an important part of the energy mix in the near term. India's existing coal- plants were relatively young at 14 years, and in FY24, 23% of India's power capacity addition came from coal. The Power Ministry recently doubled the expected thermal capacity addition to 88GW over the next 5 – 8 years.^{15,16} A December 2023 government press release clarified:

Energy security of the country cannot be achieved by renewable sources of energy alone because solar power is not available round the clock and wind energy is intermittent ...We need to add capacity that can provide round-the-clock power. Hence, dependence on coal-based generation is likely to continue till cost-effective energy storage solutions are available.¹⁷

Given the continued need for baseload power generation from coal, India had turned its attention to improving efficiency at thermal power plants and mandating emission caps. For example, it announced plans to develop the country's first 800 MW advanced ultra-supercritical (AUSC) thermal plant with indigenously developed technology. While India's thermal power plants operated at an average efficiency of 32%,^b AUSC plants operated at 46%,¹⁸ enabling savings on coal consumption.¹⁹

Policy Measures

Experts recommended implementing a mandatory carbon tax and winding down fossil fuel subsidies. In 2023, India's energy subsidies reached a 9-year high of \$39.3 billion. Of this, 50% was directed to power consumption, 40% to fossil-fuel production, and 10% to clean energy.²⁰ Fossil-fuel subsidies were largely executed in the form of import and consumer tax concessions,²¹ while subsidies on power consumption accrued to the agricultural and residential sector, with farmers receiving 75%

^b Thermal efficiency was defined as the percentage of thermal energy contained in the fuel that was converted into electrical energy that could be fed into the grid. So a 1000MW coal plant with 32% efficiency needed 3125MW of thermal input while one with a 46% efficiency needed 2174MW of thermal input. The latter therefore leads to lower coal usage and emissions.

of such subsidies. These subsidies were deeply embedded in Indian politics, with successive political parties promising free or cheap electricity to gain electoral favor.

To make up for the significant under-recoveries from supplying households and farmers, state distribution companies (DISCOMS), responsible for power transmission and distribution, charged Commercial & Industrial (C&I) users higher rates.²² This led industrial users to increasingly set up captive power plants. By 2024, 30% of India's industrial power demand was from captive production, with a total installed capacity of 78.4GW. Of this, 90% was thermal and 10% renewables.²³

Tata Power – History, Progress, and the “Net-Zero” Promise

Tata Power stood at the forefront of India's energy transition. The firm had a long history deeply intertwined with the country's development. More than 100 years ago, Indian industrialist and visionary Jamsetji Tata had a lightbulb moment at a riverside picnic. As he observed the flowing river, he got the idea to harness the power of the flowing water to generate energy. Simultaneously, during his travels in the US, he learned that to industrialize, India needed to manufacture steel, generate clean power, and build a scientific mindset. Enthused on his return from his trip, Tata incorporated the Tata Hydroelectric Power Supply Company in 1910, intending to build India's first hydropower plant on the country's western coast to supply power to Mumbai.

Over the following decades, as Mumbai's power needs increased, Tata Power built out the Trombay thermal power plant to provide an uninterrupted power supply to the city. It continued building thermal assets across North, West, and East India. While thermal power generation remained Tata Power's mainstay, the firm slowly started diversifying. It established a solar cell manufacturing facility in 1991, entered the power distribution space in Delhi in 2002, formed its renewable arm in 2007, and soon after, commissioned its first solar asset and India's first super-critical coal power plant in 2012.

Tata Power in 2024

By 2024, Tata Power was India's largest private power producer by installed capacity. It had annual revenues of \$7.4 billion, EBITDA margins of 21%, and an ROE of 12% (See **Exhibit 5a and 5b** for the company's financials). As one of India's only fully integrated private power players, Tata Power competed with other players in different segments of the power value chain.

To prepare for declining revenues from its core thermal power generation business, Tata Power began actively expanding its non-fossil-fuel businesses. This included its renewables business, which comprised utility-scale solar, wind, hybrid, and complex assets, manufacturing solar modules and cells and a solar engineering, procurement, and construction (EPC) business; its new-age energy solutions business included decentralized renewable energy (DRE) solutions such as rooftop solar and microgrids, electric vehicle charging stations, and products and services to enhance energy efficiency; and its transmission and distribution business.

On thermal power generation, its biggest competitors in the private sector were Adani Power, a private thermal power producer; JSW Energy, which operated in both thermal and renewables; Torrent Power, one of the largest players in generation, transmission, and distribution; and CESC, a utility engaged in thermal power generation, distribution, and transmission services. There were also large state-backed power producers, including NTPC, India's largest national-level power player by installed capacity and revenues, and other large state-level power producers. For clean energy, it competed with purely renewable-based private players like Adani Green and Renew Power and mixed-fuel power players like JSW Energy and NTPC. For T&D, it competed with India's largest

private T&D player, Adani Energy Solutions, and state-backed firms like the Power Grid Corporation of India and NTPC.

While the firm fared well versus peers on ESG metrics, including total emissions generated (**Exhibits 6a** for Tata Power's emissions and **6b** for a comparison with its peers) and was also the first power utility in India to have a validated SBTi target, analysts noted that its margins and stock price performance had lagged its peers (See **Exhibit 7a** for competitors' financials and **Exhibit 7b** for stock price performance for 5 years ended 2020 and from 2020 to date).

Tata Power's Clean Energy Strategy

Though Tata Power had forayed into clean energy, Dr. Praveer Sinha's appointment as CEO in 2018 catapulted the firm to the forefront of India's energy transition. An electrical engineer with over 40 years of experience in the power sector, Sinha was described as a leader with a hands-on approach, a focus on new energy solutions, and a passion for democratizing energy supply. A proponent of lifelong learning, Sinha completed a PhD in smart-grid technologies during the COVID-19 pandemic and continued to serve as a researcher at the Indian Institute of Technology, Delhi, alongside his day job.²⁴ Sinha reflected on his early days as CEO;

We had 25-year purchasing power agreements (PPAs). I could just relax and go play golf in the afternoon. Why change anything? But I had a strong conviction that we needed to reinvent Tata Power, bring in Tata Power 2.0, and prepare the company for the next 100 years. We needed to shift focus to the new energy space. Clean energy is quite simply good business. We are living on borrowed time. Soon, policymakers will come after us hammer and tongs, imposing deadlines and penalties. It's already happening in Europe. We need to take preemptive action. Besides, the Tata Group has always believed in making profits with purpose.²⁵

Under Sinha, Tata Power announced a net-zero commitment by 2045, committing to completely phase out thermal capacity by then, subject to the completion of contractual obligations and the useful life of existing plants. Some questioned the firm's decision not to invest in thermal generation and give up cash-accretive opportunities to invest in distressed thermal assets.

For instance, Jindal Power, a subsidiary of Jindal Steel and Power Limited (JSPL), was one of India's leading power producers, focusing on thermal and hydroelectric power generation. It procured several thermal plants, offering \$1.68 billion for a distressed Indian coal-fired power plant built by KSK Energy Ventures, which developed and operated power plants.²⁶ Adani Power, another major Indian power generation company and part of the Adani Group, wanted to expand its thermal power capacity and diversify its operations. It acquired Coastal Energen for ~\$397 million in 2024.²⁷ Both companies were also racing to acquire the distressed and debt-laden 1,350 MW Sinnar Thermal Power Ltd in Maharashtra.²⁸ (See **Exhibit 8** for past acquisition of distressed thermal power plants by key players)

Renewables

Renewables presented the biggest opportunity for Tata Power in India, and Sinha was optimistic about it. He explained, "Despite good progress, India's entire renewable capacity addition in 2023 was only 6% of China's. We have a long way to go; Tata Power aims to be an integral part of this journey."²⁹

Increasing Renewable Energy Capacity

In 2022, the firm launched a new platform, Tata Power Renewable Energy (TPRE), with a \$525 million investment from Blackrock and Abu Dhabi's Mubadala Investment Company for a 10.53% stake to further its renewable ambitions. Through this entity, Tata Power planned to invest the majority of its planned ~\$2.5 billion annual capital expenditures in renewables, aiming to increase renewable capacity from 5GW in 2024 to 30GW by 2030 and the share of clean energy in its portfolio from 40% to 70%. (See **Exhibit 9** for Tata Power's Capex distribution and **Exhibit 10** for installed capacity of power generation sources and the trend of electricity production from clean energy).

The firm faced several challenges to meeting this target, of which land acquisition was arguably the biggest. Renewable energy build-out required large swathes of barren real estate in specific locations. Power output per square unit of land mass was far lower for renewables than for thermal capacity, given the intermittency of its availability. As land acquisition was under the control of both the center and state governments, developers needed to navigate multiple state-specific procedures, approvals, and complexities – including the absence of clear land records and titles, inflated land values in areas with high solar potential, and local conflicts – which frequently derailed project execution.

Insufficient connectivity was also a major challenge. Solar plants were often completed before the lines needed to connect them to the nearest grid substations were installed. There was also a serious shortage of substations, particularly in areas abundant in renewable energy resources. Additionally, renewable energy installations were concentrated in India's south and western states – 7 states accounted for more than 70% of India's RE capacity – while India's power deficit states were located in its northeast. Though the Power Ministry waived inter-state transmission charges to address this imbalance, transmission was still challenging.³⁰

Tata Power sought to capitalize on the growing demand for reliable, round-the-clock renewable energy by increasing its participation in hybrid solar-wind-storage projects. As buyers moved beyond simply purchasing power to seeking comprehensive energy solutions that offer 24/7 reliability and resilience, Tata Power positioned itself to meet this need. Sinha explained, "Buyers want comprehensive energy solutions, 24x7 reliability, resilience, and security." While pure solar tariffs could settle at INR 2–2.5/ kWh, solar-wind-storage hybrids stood at INR 4–5/kWh, on par with newbuild coal plants.

While pure solar tariffs remained low, hybrid solutions, combining solar, wind, and storage, were priced competitively with coal, offering both environmental and economic benefits. India experimented with various hybrid tender formats, and 'Firm and Dispatchable Renewable Energy' (FDRE) contracts emerged as a preferred choice, ensuring power delivery according to a fixed schedule. Tata Power secured significant contracts, including a 966 MW solar-wind hybrid plant for Tata Steel and a 1.3 GW FDRE deal with SJVN, a public sector hydroelectric power generation company. Sanjeev Churiwala, appointed CFO in 2022, explained;

Pure solar/ wind contracts have become commoditized. Their 13-15% returns are insufficient. And risks associated with pure solar contracts are high. We rely on China for solar panels, and there is significant uncertainty on land acquisition and access to transmission lines. Players that bid aggressively in this segment are going belly-up. We stopped bidding for those projects and moved to hybrid tenders that give us a high teens return. To us, returns and cash flow matter more than the top line. Our minimum internal returns threshold is 18%. With these returns, I can allocate greater CapEx to renewables.³¹

Production of Renewable Components

Tata Power's renewable strategy was centered on building a fully integrated end-to-end value chain both to gain better control over its own supply chain and to further India's goals of energy security. As solar modules constituted ~60% of the cost of setting up a solar plant, Sinha believed that in-house module manufacturing was a natural extension of its renewable business.

Like other countries, India relied heavily on imports to meet its solar module requirements – it imported two-thirds of its solar cell and 100% of its wafer requirements since 2021.^{32,33} China dominated solar module manufacturing, particularly in upstream components, but India was expected to gain market share. India had introduced several measures to boost local production, including a \$2.8 billion incentive scheme for the domestic production of modules.³⁴ These efforts had yielded results, and by 2024, India had a solar module manufacturing capacity of 65.5GW, up from 2.3 GW in 2013,^{35,36} and a cell manufacturing capacity of ~6GW, though it still had low capabilities in the upstream segments of the value chain.³⁷ However, demand for solar module and cell manufacturing in India was projected to grow rapidly, with projections estimating that India would need a total capacity of 110 GW by 2030 to meet its solar energy goals.³⁸

Tata Power was an important player in this journey. In late 2023, it commissioned a greenfield investment in a 4.3 GW solar cell and module plant in Tamil Nadu. This was built on the firm's existing 530MW of cell and 682MW of module manufacturing capacity in Bengaluru. Sinha was thrilled; "The commencement of production is a significant step towards indigenization in the solar value chain and achieving self-sufficiency. We believe that this plant will lead the way in supporting the country's vision for a Net-Zero carbon future."³⁹ He added, "We can now control costs and the quality and consistency of supply. Better quality enables us to charge a 10% - 15% premium. Domain knowledge is acquired over time. We learn and re-learn as we want to get each step right. This is not arrogance; it is reality. We have no real competition; they don't have our DNA."⁴⁰

While some experts were optimistic about Tata Power's investments in module manufacturing, others noted that India would never be able to compete with China in terms of price. China's modules were priced at \$0.15/Watt vs. \$0.22 in India, \$0.30 in Europe, and \$0.40 in the US. Given China's significant overcapacity – its total module production capacity reached 1000 GW/year in 2023, more than double global demand, which was expected to reach 980 GW/year by 2035 – experts expected continued downward pressure on module prices. In January 2024, JSW Energy put its solar module manufacturing plans on hold, citing a record fall in solar panel prices. But Sinha remained unphased, explaining that overcapacity issues would not impact the firm's business given government restrictions on using imported modules in projects.

Other Non-Fossil Fuels: Nuclear and Green Hydrogen

Two other key priorities in Tata Power's non-thermal energy generation were nuclear and green hydrogen (hydrogen produced via electrolysis of water using renewable energy). Given India's lack of natural gas resources, nuclear was envisioned as an important baseload power source.⁴¹ Sinha explained, " _ If one system fails, you need a backup. For resilience, we need to build nuclear capabilities."⁴²

India's state-backed Nuclear Power Corporation of India Limited (NPCIL) had been the primary nuclear energy developer. However, a recent opening up of the industry to the private sector presented opportunities for Tata Power. In early 2024, the government announced plans to partner with the private sector on the development of ancillary infrastructure for large-scale nuclear power plants and on "Bharat Small Reactors" (BSRs), compact nuclear reactors modeled on India's 220 MW pressurized heavy water reactor technology. Tata Power partnered with the Department of Atomic Energy on the

latter. Critics noted that although these reactors occupied less space and could provide power in remote regions, they were expensive to operate on a per-unit basis.⁴³

The company also looked towards green hydrogen production using renewable energy.⁴⁴ India had recently launched a Green Hydrogen Mission with incentives for its production, but, Churiwala explained that there was a wide gap between theory and reality; “Building hydrogen production capabilities requires systems thinking; a building out of complete infrastructure for its storage and transport. We have a hydrogen mission, but execution is in a very confused state.”⁴⁵ Given this backdrop, Tata Power focused on producing and selling renewable energy to green hydrogen developers, which constituted 70% of the cost of green hydrogen production.

New-Age Energy Solutions

Decentralized Renewable Energy (DRE) Solutions: Solar Rooftop, Microgrids, Solar Pumps

Sinha was deeply passionate about democratizing access to energy and was keen to see Tata Power drive decentralized renewable energy (DRE) solutions to tackle India's fragmented power consumption. Despite India's near-total household electrification, power supply remained unreliable, especially for low-consumption users in rural and urban settings.⁴⁶ The cost to supply power to low-consuming users, particularly in remote areas, was often higher than expected returns. Tata Power's approach focused on a few solutions:

Roof Top Solar (RTS) emerged as a key part of this strategy, offering scalable solutions from small home units to larger community installations. By 2024, Tata Power had a 13% market share in RTS, with 2GW installed across 70,000 customers. Despite the potential for over 250 million households to adopt RTS, uptake remained limited due to low consumer awareness, financing challenges, and difficulties with DISCOMS being understaffed and lagging in technology. Tata Power aimed to expand its market share to 20%, driven by innovative financing and service models tailored to residential, commercial, and industrial users.

By early 2024, 75% of India's 14.5GW RTS capacity came from the C&I sector, though they sourced only 3.5% of their power demand from it, leaving room for growth. Despite the potential for 250 million households to deploy 637GW, residential RTS uptake was limited.^{47,48,49} Sinha explained;

We want to empower rural and urban communities to produce their own energy and sell it to the grid. Our solar rooftop EPC project targets these communities. Earlier, teasellers had no light after 6 pm; local doctors could not see patients in the evening. Our facilities have enabled an entire ecosystem change, a socio-economic transformation.⁵⁰

Churiwala elaborated, “Akin to any consumer business, this segment is about creating the right channel partners. We have built a strong network over the years, but given our growth trajectory, we need to tie up with more retailers, distribution channel partners, and banks.”⁵¹

Microgrids, or self-contained electrical networks, typically constituting solar panels and battery systems, were also important for Tata Power's plan to increase access to power in remote areas. The company installed 200 microgrids across small towns and villages in North India, impacting over 22,000 people. Despite being loss-making, it was viewed as a long-term, scalable solution to reduce diesel dependency and emissions. Sinha envisioned these efforts as a model for India and sub-Saharan Africa, combining technological innovation and sustainability. As of 2024, Tata Power was at the forefront of India's nascent microgrid landscape. He described;

Our microgrid program is the largest in the world and we are a leader in universal energy access. It depends on technological interventions and product innovation to bring low-cost solutions to rural areas in a scalable manner. We had MIT engineers working on a 20-watt fan, refrigerator, and a 30-watt TV set. Everything in the project was an innovation; nothing could be bought off the shelf. Our vision is to empower a billion lives through sustainable, affordable, and innovative energy solutions. This is part of our commercial strategy, not a Corporate Social Responsibility Initiative.⁵²

Agricultural Solutions, particularly access to solar water pumps, were another focus for Tata Power's energy access initiative. Agriculture in India employed 54.6% of India's population but contributed only 18.6% to GDP and faced yield declines of 10-20% due to fragmented land holdings and outdated technologies leading to lower productivity. In some states, it consumed more than 30% of power demand. With a 30% market share, Tata Power had sold over 100,000 solar pumps, supporting the government's welfare program to provide farmers with incentives to solarize pumps. These efforts aimed to address water security and reduce power demand in the agricultural sector.⁵³

Electric Vehicle (EV) charging.

Tata Power was keen to participate in India's target of 30% adoption of EVs by 2030. Several demand—and supply-side incentive schemes to accelerate production and uptake led to a plethora of activity and entrepreneurship in electric mobility across vehicles, batteries, and charging infrastructure while EV sales were growing fast.

The company supported the group company Tata Motors, a leader in EVs, via home charging kits and building out public infrastructure. As of 2024, it was a market leader in charging infrastructure with a 60% market share. It established 5,400+ public car and 850 bus charging stations across 530 cities and towns and 86,000 home charging units. It had collaborated with several government and private bodies to set up EV charging solutions, including Indian Oil Corporation Limited (ICOL), to roll out ultra-fast EV charging infrastructure across IOCL retail outlets and Zoom Car, an Indian car-sharing platform. It was exploring opportunities in the electric 3-wheeler and 2-wheeler charging market, India's two biggest EV segments by sales. It targeted installing 750,000 home charging units and 10,000+ public EV charging points within the next few years.

Energy "Smartness:" Efficiency via Smart Meters, Home Automation

Tata Power saw a significant opportunity in the market for energy efficiency. Improving power consumption efficiency first relied on better availability and data tracking. By 2024, India had one of the largest power grids in the world, but it mostly relied on one-way communication using basic meters to set fixed electricity prices. The government's National Smart Grid Mission aimed to modernize the grid to allow two-way communication of both energy and information. This would involve installing smart meters, sensors, and automated systems to access real-time data. They would also allow for flexible pricing, adjusting rates depending on the time of day to manage demand and supply more efficiently.^{c,54,55,56} The project was under pilot, but progress had been slow, with only 8 million smart meters installed by the end of 2023, far short of the 250 million goal by 2025 due to high costs and resource challenges.

^c India's Power Ministry had announced ToD tariffs for C&I users effective April 1, 2024, and other consumers (except farmers) effective April 1, 2025. For C&I users, peak hour tariffs would be at least 1.20x the regular rate, and for other users 1.10x, while tariffs for solar hours would be at least 20% lower than regular rates

Tata Power was keen to contribute to India's grid modernization journey and had rolled out more than 1 million smart meters in states where it controlled distribution. It had also recently undertaken a smart metering project for one of the state-owned state power distribution companies.

The second element of improving energy efficiency was introducing smart energy management technologies in appliances and urban infrastructure to help users optimize usage.⁵⁷ Tata Power saw this as a big business opportunity. Its new energy business launched IoT-based smart energy solutions, including energy-efficient sensors and lights, which enabled users to save up to 70% on their power bills. It introduced an Energy-as-a-Service offering for C&I clients, which enabled them to monitor consumption and implement cost and emissions-saving strategies. Sinha explained, "We need to bring smartness at the grid and consumer level. Smart technology can be used in commercial and residential buildings to enable users to optimize and reduce cost, and utilities conserve energy. There is room for neither famine nor feast in the power industry."⁵⁸

Storage and Integrated Solutions/Other Solutions

Increasing Investments in Storage Solutions

Renewable energy's temporal and output variability made grid integration complex, as generation often deviated from demand and forecasts. To meet the demand for round-the-clock renewable energy, Tata Power was increasingly investing in storage solutions, including Pumped Storage Projects (PSP) and Battery-Energy-Storage-Solutions (BESS).⁵⁹ Although India had significant pumped storage potential, PSP projects were generally difficult to execute, given challenges with land acquisition. Sinha explained, "Greenfield hydro projects are difficult to execute given environmental concerns and issues with risk and regulations. Besides, many high-potential areas like those in the northeast are inaccessible. But our pumped hydro projects are brownfield; we already have the reservoirs, land, and infrastructure. That gives us a competitive edge; our capex cost per megawatt is lower than others."⁶⁰

The firm had recently signed an agreement with the Maharashtra state government to develop two large-scale pumped-storage plants. The BESS landscape was fast evolving as developers raced to find the lowest-cost, medium-duration storage medium. Sinha explained, "On BESS, the disruption has to come from new chemistry and technology. Everyone knows that rare metals are rare. We need to use metals that are available in abundance; use organic materials."⁶¹ A range of incentives had catalyzed significant activity in the space, and Tata Power, too, was making strides.⁶² It had recently won a tender to build India's largest solar- BESS project and had contributed \$100 million to fund R&D into new and emerging materials and technologies at MIT.

Providing Integrated Power Solutions

A key strategic objective for Tata Power was introducing new ways to meet customer energy needs. One approach involved selling power directly to consumers through agreements bypassing traditional state-run DISCOMs. Another goal was to expand its role in short-term energy markets, such as the Green-Term Ahead market. By 2024, 85% of power in India was traded through long-term agreements between state utilities and power producers, typically lasting 25 years, while only 15% was traded in short-term markets. In comparison, in more developed markets, up to 80% of power was traded in short-term or spot markets. Although long-term contracts provided stable returns for developers, they reduced flexibility in integrating renewable energy into the grid.

This structure also contributed to DISCOMs' inability to meet centrally mandated Renewable Purchase Obligation (RPO) targets^d as existing long-term PPAs kept them locked into thermal power purchases. As India's power market transitioned, Tata Power's renewable arm hoped to benefit from increased volume and liquidity in wholesale spot markets.

The Transmission and Distribution (T&D) Business

Expanding its T&D business was a critical strategic priority for Tata Power, to address inefficiencies in India's power grid. As of 2024, India's power distribution was highly inefficient.⁶³ The grid suffered from high transmission losses due to outdated infrastructure and rampant malpractices. Most state-run DISCOMs did not have the financial wherewithal to invest in upgrades.

The government's Revamped Distribution Sector Scheme aimed to reduce these losses and close the revenue-cost gap through smart meters and infrastructure improvements.⁶⁴ While this vision marked a step in the right direction, many believed that its execution would rely on the privatization of power distribution. Private operators had successfully lowered Aggregate Technical and Commercial (AT&C) losses in states where they controlled power distribution.^{e,65,66} They used a range of strategies, including working with users to create a mindset of electricity as a paid service, improving metering and billing practices, and incorporating AI and data analytics to better match power demand with supply.^{67,68} (See **Exhibit 11** for AT&C losses of private operators)

Tata Power played a key role in reducing losses and improving billing through advanced technologies and data analytics. Its T&D portfolio encompassed over 6,000 circuit kilometers of transmission lines (under construction) covering six states and distribution operations to 12.5 million customers. The firm showed great success in improving distribution efficiency, including installing smart meters and forming self-help groups to raise collection rates. Sinha viewed this segment not only as a steady future revenue line but also as a show of commitment toward the government's grid modernization mission. He noted, "It can be challenging in some remote areas, but it gives us an assured return of ~14% - 16%. We understand this segment well." Chuirwalla echoed the appeal, stating, "T&D could look better than renewables on a risk-adjusted basis. It is a low capex, high returns, high free cash flow business, and as a CFO, I'm always excited by this." Tata Power expected continued privatization of the power sector and aimed to expand its distribution network to 40 million customers by 2030 through public-private partnerships and new technologies. Analysts noted that DISCOM privatization was a "political economy issue" and that "there has been a lot of resistance from states,"⁶⁹ Sinha was optimistic, stating, "That's the last part of the power sector changes or transition that is expected. We are very confident that. . . [this] will happen."⁷⁰

Commitment to Sustainability

Tata Power closely tracked its sustainability progress through clear, measurable targets. To achieve net zero by 2045, it set interim goals, such as reducing greenhouse gas emissions by FY27. By FY24, Tata Power had already shifted 40% of its portfolio to clean energy, with plans to increase this to 70% by FY30 and reach 100% by 2045. "These targets were tied to specific metrics, including water

^d Buyers could purchase RECs to fulfill these obligations. Recently, the government had introduced Energy Storage Obligations (ESO) as a separate component of the RPOs, provided that 85% of energy stored in the ESS was procured from RE.

^e Under the license model, distribution was privately owned and operated; under the franchise model, distribution was state-owned but operated by private firms. Franchised operators were not included in tariff setting and thus tended to invest less.

neutrality, net-zero waste, and biodiversity protection by 2030," said Himat Tewari, Chief Human Resource Officer.

The company incorporated these sustainability metrics into its annual business plans, ensuring each business unit was accountable for both financial and sustainability targets. Vaishnavi Prabhakaran, Head of sustainability, explained that the goals were closely tied to employee performance, "It goes into the KPIs of people. Business units were not only measured on traditional financial success but also on their progress in reducing emissions and advancing clean energy projects."

In addition to environmental goals, Tata Power also worked on social sustainability, particularly through workforce transition initiatives. Project Daksha aimed to retrain employees from traditional thermal power roles to new positions in renewable energy, ensuring they remained relevant in the company's future operations. Its AI-based Climate Group also encouraged its 23,000 employees to adopt sustainable habits, such as using public transport and reducing waste. This initiative aimed to expand to one million participants by engaging communities where Tata Power operated.

Tata Power's sustainability tracking was not just driven by regulatory requirements; the company voluntarily adopted this method in FY22, ahead of SEBI's^f mandatory disclosure requirements for top companies. "We have always aimed to go beyond compliance," Tewari said.

The Path Ahead

As India's large private and public power players continued to bid for distressed thermal assets and invest in new thermal capacity, Sinha reflected:

We were the first Indian utility to announce an end to new thermal investments. I knew this would be controversial. Most of our employees have worked only in coal or hydro. Distressed thermal assets offer easy returns. We acquired one in December 2019 and made a 35% return. I am advised to do more of this. . We need to consider the long-term, and invest in smarter and newer solutions. 24x7 renewable power is possible; the technologies exist. [For competitors investing in coal] You cannot stop them from jumping into the Sea. You can delay writing the obituary, but the obituary has to be written.⁷¹

Churiwalla added, "Regulated returns on thermal assets are still a high 16% - 17%. Currently, there is no comparison of what thermal can bring us in terms of returns and cash. We have compromised some short-term returns to meet our climate agenda. But, our investors can sleep well at night if they are focused on the long-term. I'm not sure that is the case for our competitors."⁷² He added that competitors' high capex figures were inflated by high leverage and faced stranded asset risk by continuing to invest in thermal. With a range of Tata Power's capex investments likely to convert into revenues soon, he was confident of an improvement in the firm's return and cash flow profile.

Some also questioned Tata Power's presence across numerous business segments, describing its strategy as "confused." They questioned the scale and profitability of these channels, recommending that the firm narrow its focus down to a few core competencies. While Sinha agreed that running a fully diversified power business presented its challenges, he disagreed with the assessment, stating;

The power sector can no longer be viewed in silos. The lines between generation, transmission, and distribution are blurring. Generation can now take place at the

^f The Securities and Exchange Board of India (SEBI) was the Indian securities market regulator.

household or community level. Technology has enabled a bi-directional flow of data and energy. We can now monitor energy generation and consumption in real-time and reduce inefficiencies. If I can better anticipate demand, I can better plan capacity. Hence, having end-to-end infrastructure management capabilities is ever more important. The power sector is going through a transition from a monopolistic, regulated, take-it-or-leave-it model to a customer-centric model. The uniqueness of Tata Power is that we are a fully integrated power company; we have a full line of sight right up to the consumer.⁷³

That said, Sinha acknowledged the concern that Tata Power's transition was fundamentally altering its business model from that of a regulated entity with assured long-term returns to a customer-centric business with variable revenues and returns. He reflected, "Today, 50% - 60% of our returns come from the regulated business, down from 95% a few years ago; the rest is market-driven. Classic utility investors prefer assured returns, but our mix has undeniably changed."⁷⁴ Himat Tewari, Sustainability Head at Tata Power, elaborated;

The transition marks a deep domain transformation of our business, structure, people, and interaction with the community. The classic B2G business model is morphing into a B2B and B2C business; from an electron provider to a solutions provider, from predictability to unpredictability. We have to rehabilitate, reskill, and upskill employees to manage this transition with dignity.⁷⁵

As Sinha and Churiwalla sat down to discuss the 5-year capex plan and strategy, several questions arose. Sinha was focused on technological change and wondered how aggressively the firm should bet on new and emerging technologies and business lines. He reflected;

We need to constantly learn and evolve. We need to be conservative yet aggressive. Being too conservative means missing out on opportunities. But if you want to run a marathon, you need to be physically prepared. We have a fiduciary responsibility to our shareholders. We need to be cautious and calibrated.⁷⁶

Meanwhile, Churiwalla's concern was budget allocation. With limited capital and competing demands, he needed to ensure they were allocating capital wisely. He said;

The business lines need to give us cash, then we will allocate capex. We need to look at risk-adjusted returns. Last year, we had to cut off bidding for pure wind and solar. We need ~ 19% returns. Our competitors are focused on capex and debt-driven top-line growth. We don't need growth for growth's sake. Capital allocation is a rigorous process that starts at the top.⁷⁷

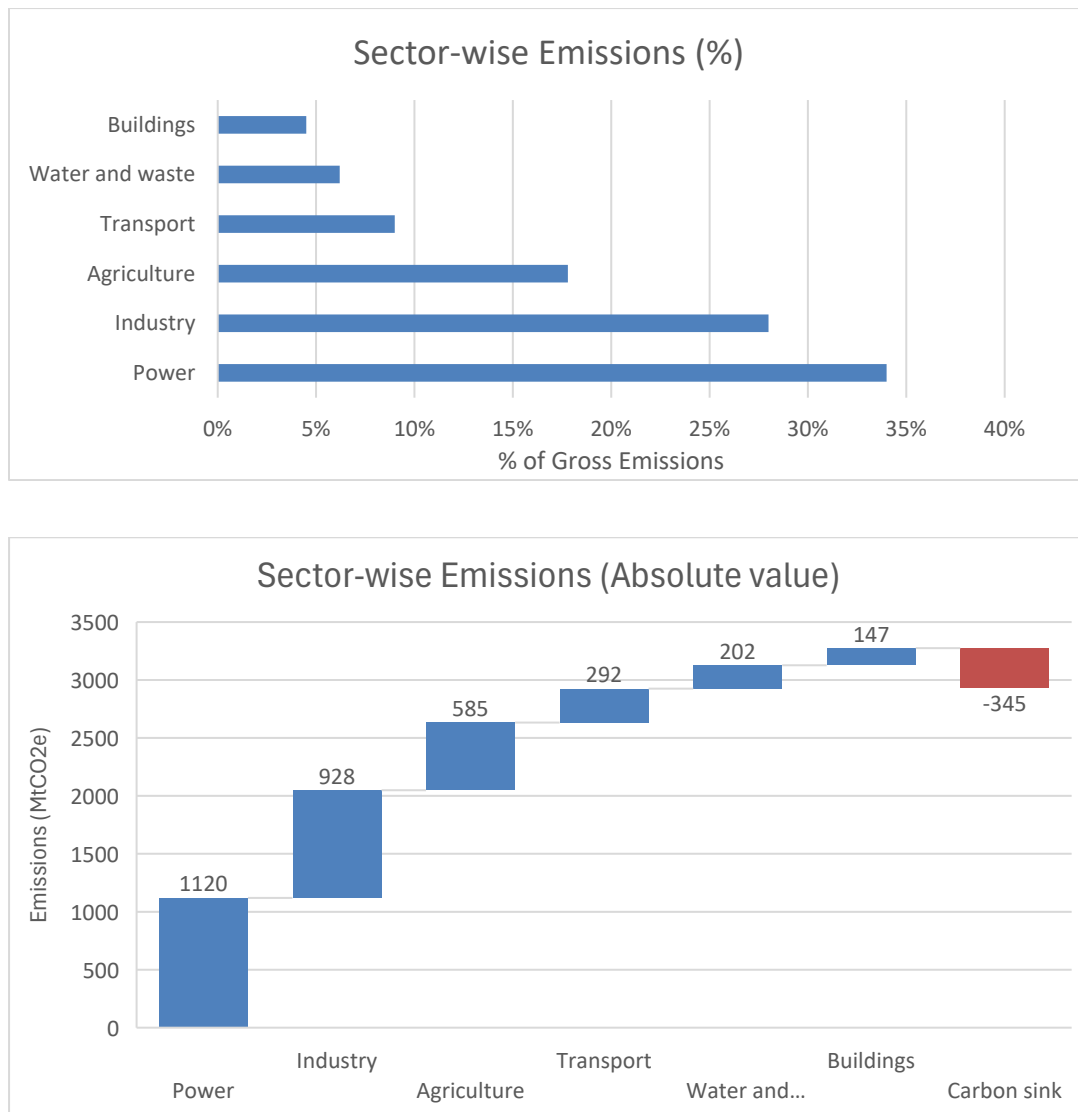
At the core of Tata Power's decision on capital allocation was the question of how the firm should navigate the evolving opportunities and risks presented by India's energy transition. By being an early mover, was Tata Power positioning itself well to take advantage of emerging opportunities and safeguard the firm's assets and long-term returns against climate-related risk, or did it represent a compromise on financial returns? Was the firm involved with too many disparate and financially unviable business lines, or was this experimentation necessary to pave the way to the Tata Power 2.0 of the future?

Exhibit 1 Overview of Elements of the Power Sector.

Processes	Definition	Key Players in India
Generation	This is the process of producing electricity by converting raw energy sources such as coal, natural gas, nuclear energy, or renewables like wind, solar, and hydropower into electricity. Power plants operate based on different technologies: thermal power plants burn fuel to generate heat, nuclear plants use nuclear fission, and renewable plants harness natural energy (e.g., solar panels capturing sunlight or wind turbines using wind).	Power generation companies like Tata Power and NTPC, along with independent power producers (IPPs), and renewable energy developers such as ReNew Power. These companies manage large-scale production facilities to generate electricity.
Transmission	Once electricity is generated, it needs to be transported over long distances to reach the areas where it's needed. This involves high-voltage transmission lines supported by tall towers. Transmission lines connect power plants to substations. To minimize energy loss during transmission, the electricity is carried at very high voltages, and transformers are used to step up the voltage before transmission and step it down when it reaches its destination.	Power Grid Corporation of India manages most of India's interstate transmission system. State Transmission Utilities (STUs) manage intra-state transmission.
Distribution	After electricity reaches substations through transmission lines, it must be distributed to homes, businesses, and industries at lower voltages. Distribution companies step down the voltage to safer levels and deliver it to consumers through a network of lower-voltage lines. This stage also includes metering, customer connections, and ensuring a reliable power supply.	Distribution companies (DISCOMs) handle retail electricity supply. Examples include BSES, Tata Power-DDL, and other state-run utilities. They are responsible for the last-mile connection and managing customer billing and services.

Source:

Created by Aindrita Sinha, Research Associate, HBS India Research Center, based on https://www.energy.gov/sites/default/files/2023-11/FINAL_CESER%20Electricity%20Grid%20Backgrounder_508.pdf, accessed October 2024.

Exhibit 2 India: Baseline Emissions in 2019.

Source: Adapted by casewriters from Rajat Gupta, Shirish Sankhe, Naveen Unni, and Divy Malik, "Decarbonising India: Charting a pathway for sustainable growth," McKinsey Sustainability, October 2022, <https://www.mckinsey.com/~/media/mckinsey/business%20functions/sustainability/our%20insights/decarbonizing%20india%20charting%20a%20pathway%20for%20sustainable%20growth/Decarbonising-India-Charting-a-pathway-for-sustainable-growth-ES-Oct-2022.pdf>, accessed October 2024.

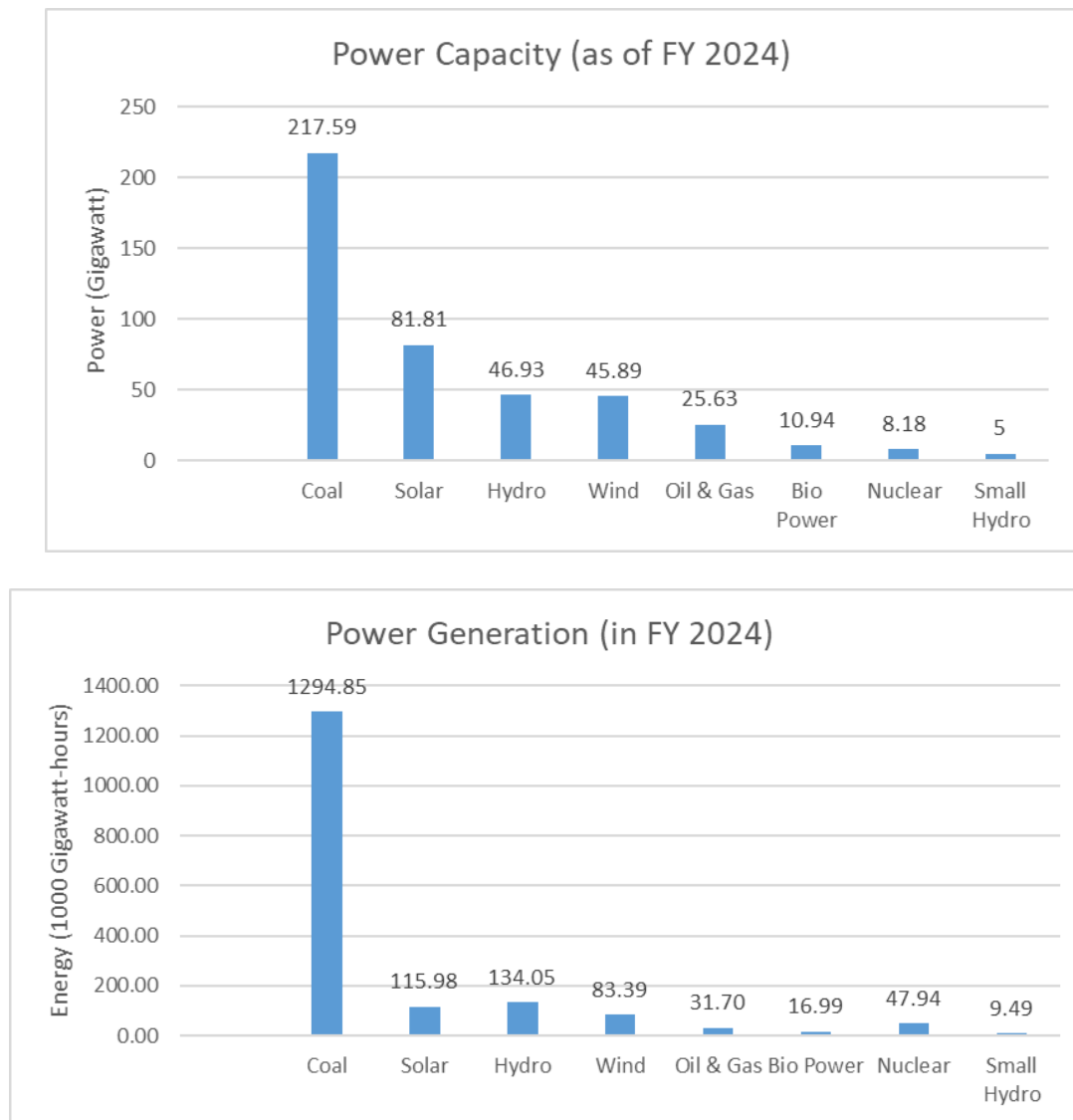
Note: MtCO₂e stands for Metric Tons of Carbon Dioxide Equivalent. It is a standardized measure used to express the total greenhouse gas (GHG) emissions in terms of the amount of carbon dioxide (CO₂) that would have the same global warming potential (GWP)

Exhibit 3 Current versus Forecasted Installed Power Capacity

	Current (May, 2024)		Expected (FY 2032)	
Power Source	Installed Capacity (GW)	% of Total Capacity	Installed Capacity (GW)	% of Total Capacity
Coal	217.6	49%	259.6	29%
Oil and Gas	25.4	6%	24.8	3%
Total Fossil Fuel	243	55%	284.5	32%
Large Hydro	46.9	11%	62.2	7%
Small Hydro	5	1%	5.5	1%
Solar	84.3	19%	364.6	40%
Wind	46.4	10%	121.9	14%
Bio Power	11	2%	15.5	2%
PSP	4.7	1%	26.7	3%
Nuclear	8.2	2%	19.7	2%
Total Non-Fossil Fuel	201.8	45%	616	68%
Total Capacity	444.8	100%	900.5	100%

Source: Adapted by casewriters from India Climate and Energy Dashboard. "Home." <https://iced.niti.gov.in>, accessed November 2024. Ministry of Power, Government of India. "132 GW of electricity capacity under construction, 517 GW capacity anticipated to be added by 2031-2032: Union Power and New & Renewable Energy Minister." Press release, December 15, 2023. PIB Delhi, <https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1986616>, accessed November 2024.

Note: As of May 2024, the private sector managed 52% of the total installed capacity.

Exhibit 4 Installed Power Capacity and Power Generation by Source as of August 2024.

Source: Adapted by casewriters from India Climate and Energy Dashboard. "Electricity." <https://iced.niti.gov.in/energy/electricity/generation>, accessed October 2024.

Note: Power capacity refers to the maximum amount of electricity a power plant or generator can produce per second, typically measured in watts. The power capacity added in FY 2024 was 26 Gigawatt. Power generation depends on both the plant's power capacity and the duration of operation. While power capacity reflects the potential energy output per second, power generation quantifies the total energy produced based on how long the plant operates at its capacity. Power generation is measured in watt-hours.

Exhibit 5A Tata Power Financials from FY2016-2019 and FY2020 to 2024. (\$ billions))

Account	FY2016	FY2017	FY2018	FY2019
Revenues	3.56	3.33	3.24	3.61
EBITDA	0.86	0.73	0.77	0.83
Margin	24.10%	21.80%	23.60%	22.90%
PAT	0.08	0.11	0.29	0.28
Margin	2.20%	3.20%	9.00%	7.90%
Total Gross Debt	4.69	5.90	5.87	5.86
Equity	1.88	2.08	2.28	2.47
Capex	0.26	0.40	0.43	0.43
Ratios				
Capex/ Sales				
Debt/ Equity	2.5	2.8	2.6	2.4
Debt/ EBITDA	5.5	8.1	7.7	7.1
ROE	4.80%	6.70%	14.80%	13.90%

Account	FY2020	FY2021	FY2022	FY2023	FY2024
Revenues	3.50	4.01	5.14	6.77	7.43
EBITDA	1.00	0.96	0.99	1.22	1.53
Margin	29%	24%	19%	18%	21%
PAT	0.16	0.17	0.26	0.46	0.52
Margin	5%	4%	5%	7%	7%
Equity	1.84	3.05	3.14	4.13	4.63
Capex	0.27	0.38	0.88	0.92	1.61
Financial Ratios					
Capex/ Sales	8%	9%	17%	14%	22%
Debt/ Equity	3.2	1.7	1.9	1.4	1.3
Debt/ EBITDA	5.9	5.5	5.9	4.9	3.9
ROE	8.20%	6.30%	10.10%	12.30%	11.50%

Source: Tata Power Annual Reports.

Note: Conversion rate used: 1 USD=82.79 INR

Exhibit 5B Tata Power Financials by Business Segment for FY2024.

Business Segment	Revenue		EBITDA		PAT	
	Amount (\$ million)	% Share	Amount (\$ million)	% Share	Amount (\$ million)	% Share
Thermal and Hydro Generation, Coal and Shipping	2369.13	32%	456.46	30%	255.10	49%
Renewables	1229.01	17%	413.33	27%	90.35	17%
Transmission & Distribution	4373.35	59%	550.31	36%	175.99	34%
Eliminations/ Others	-537.99		114.02		-4.47	
Total	7433.51		1534.12		516.97	

Particulars	Operational Income (\$ million)		EBITDA (\$ million)		PAT (\$ million)	
	FY2024	FY2023	FY2024	FY2023	FY2024	FY2023
Thermal Generation, Coal and Hydro						
Maithon Power Limited (MPL) *	405.85	365.74	104.48	93.85	54.23	40.95
Traditional Generation (incl. Mumbai and Hydro)	710.23	831.50	152.55	110.88	89.75	69.33
IEL	-	-	-	-	10.75	10.39
PPGCL	-	-	-	-	8.09	6.88
Others	15.82	15.70	17.27	16.79	18.00	21.50
Mundra, Coal and Shipping	1237.23	986.71	182.15	129.61	69.69	291.70
Total	2369.13	2199.66	456.46	351.13	250.51	440.75
Renewables						
RE Gencos	413.82	387.37	359.22	345.69	70.42	71.51
Solar EPC (TPSSL)	1416.35	830.54	86.97	62.93	47.23	26.81
Others (Incl. eliminations#)	-601.16	-227.81	-32.85	-21.74	-21.74	-15.22
Total	1229.01	990.10	413.33	386.88	102.55	88.17
T&D						
Transmission						
Mumbai	152.31	134.19	109.43	96.51	38.41	36.60
Powerlinks**	0.00	0.00	0.00	0.00	5.07	4.95
Distribution and Services						
Mumbai	530.02	600.56	76.70	65.95	25.24	22.10
Odisha	2226.60	1988.16	184.32	134.19	37.20	30.56
Delhi	1123.81	1158.84	161.49	157.75	54.72	53.15
Others	340.62	289.29	18.36	6.88	15.34	3.74
Total	4373.35	4171.04	550.31	461.29	175.99	151.11

Source: Created by Aindrila Sinha, Research Associate, HBS India Research Center, based on <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/quarterly-results-presentation/2024/Q4%20FY24%20Analyst%20Call%20Presentation-%20May%2024.pdf> (Pg 9), <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/quarterly-results-presentation/2024/Q4%20FY24%20Analyst%20Call%20Presentation-%20May%2024.pdf> (Pg 49)

Note: 1. Stake in overseas coal mines and overseas hydro assets was not considered for financial numbers related to Thermal and Hydro Generation, Coal, and Shipping.
2. Conversion rate used: 1 USD=82.79 INR

Exhibit 6A Tata Power's Emissions by Source (in MtCO₂e)

Scope 1	
Thermal - fuel and material consumption	35,637,107
Waste heat recovery, blast furnace gas	3,028,778
Hydro - fuel and material consumption	177
Clean and green generation (solar manufacturing, solar & wind O&M)	162
Corporate office	193
Smart grid technology	5,434
Total Scope 1	38,671,851
Scope 2	
Clean and green generation - Aux grid consumption	74,229
T&D - Aux consumption and losses	2,812,417
Total Scope 2	2,886,646
Scope 3	
Purchased goods and services	99,506
Waste generated in operations	143,366
Capital goods	918,441
Business travel	3,707
Fuel and fuel related activities	21,096,248
Use of sold products	2,165,486
End-of-life treatment of sold products	545
Upstream T&D	35,258
Downstream T&D	2,198
Employee commuting	36,184
Franchises	433,196
Investments	1,530,111
Total Scope 3	26,464,246
Total Emissions	68,022,743

Source: Tata Power Annual Report

Note: Scope 1 emissions are direct emissions from a company's operations, such as fuel combustion. Scope 2 emissions are indirect emissions from purchased electricity, heat, or cooling. Scope 3 emissions are all other indirect emissions in the value chain, including supply chain and product use.

Exhibit 6B Comparison of Total GHG Emissions of Tata Power and its Peers

	Tata Power	Adani Power	NTPC
Scope 1 (mil MTCO ₂ e)	38.7	72.9	352.5
Scope 2 (mil MTCO ₂ e)	2.9	N/A	N/A
Scope 3 (mil MTCO ₂ e)	26.5	N/A	N/A
Total (mil MTCO₂e)	68	72.9	355.8
GHG intensity (tCO₂e/MWh)	0.65	0.85	0.83

Source: Tata Power Annual Report

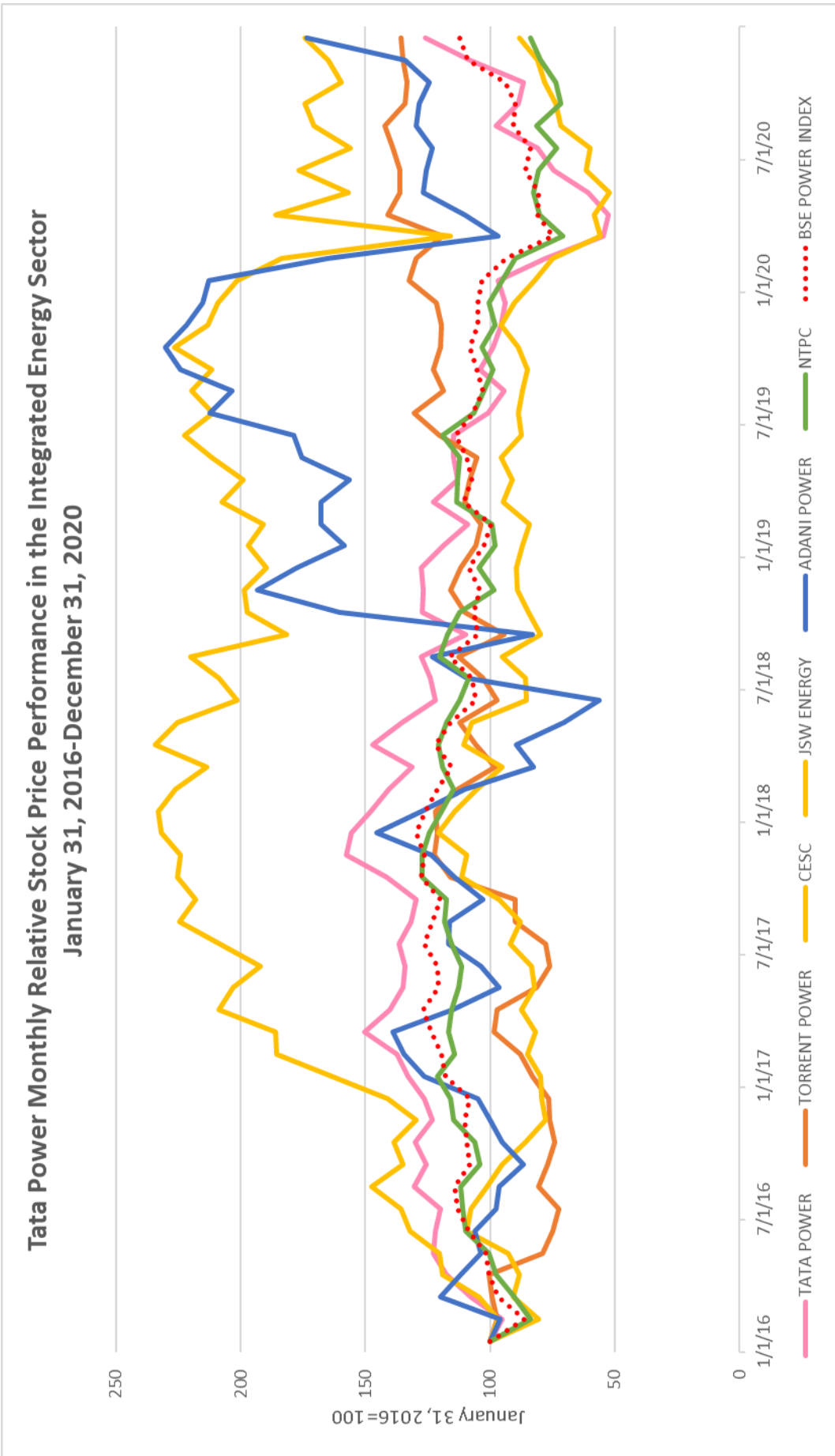
Note: GHG intensity (tCO₂e/MWh) measures the amount of greenhouse gas emissions (in metric tons of CO₂ equivalent) produced per megawatt-hour (MWh) of electricity generated. It indicates the carbon efficiency of energy production, with lower values reflecting cleaner, more sustainable energy sources.

Exhibit 7A Comparison of FY2024 Financials of Tata Power and its Peers.

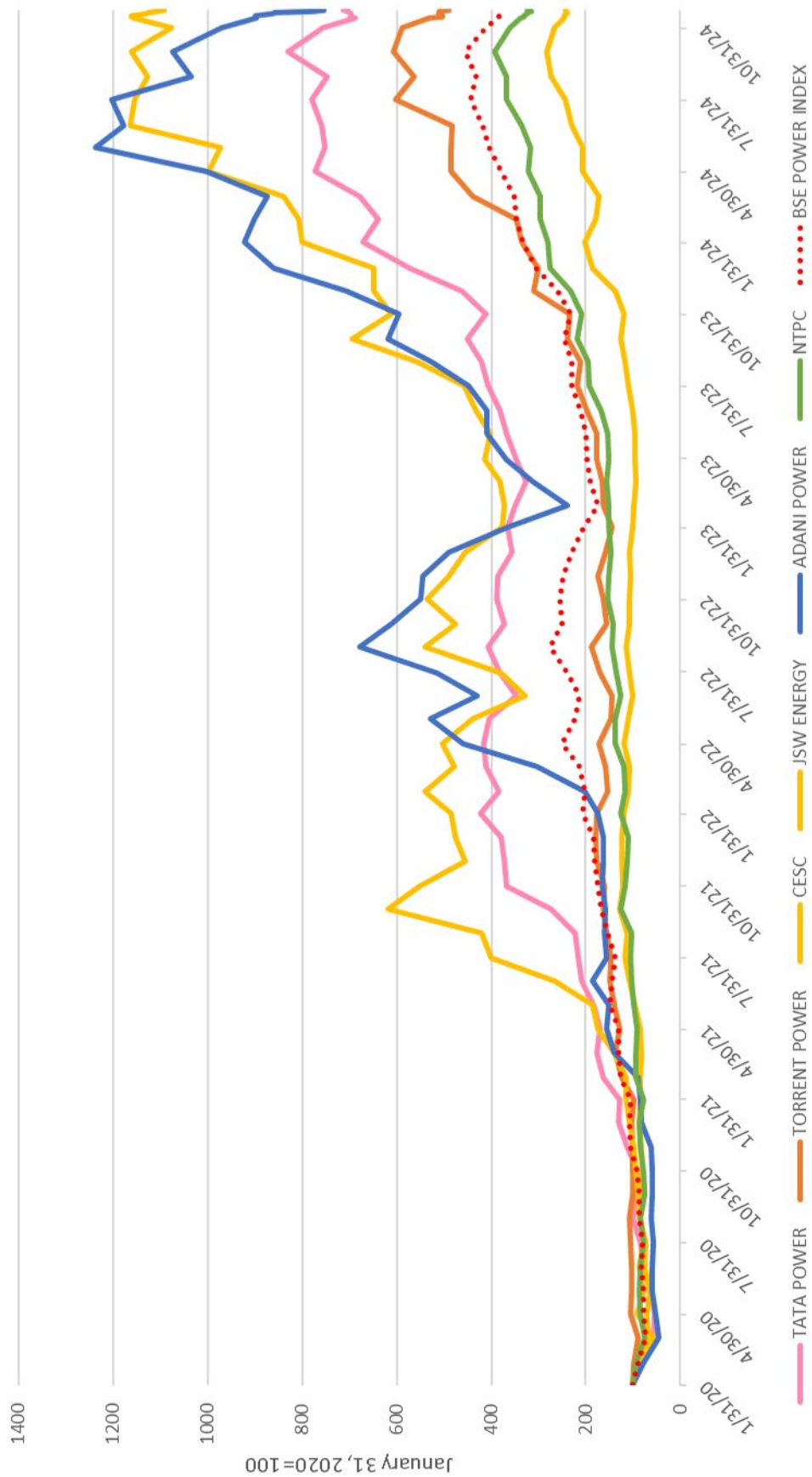
Parameters	Tata Power	Adani Power	NTPC	JSW Energy
Revenues (\$ million)	7381	7230	21728	1432
EBITDA (\$ million)	1523	3371	6643	700
Margin	21%	47%	31%	49%
PAT (\$ million)	513	2498	2558	207
Margin	7%	35%	12%	14%
Total Gross Debt (\$ million)	5991	4133	28189	3757
Equity (\$ million)	4597	5175	19804	2498
Debt/ Equity	1.3	0.8	1.4	1.5
Debt/ EBITDA	3.9	1.2	4.2	5.4
Market Cap (\$ billion)	16.5	29.4	46.7	15.4
P/E Ratio	39.7	15.6	18.2	62.6
Return on Equity (ROE)	12%	48%	12.52%	8.27%

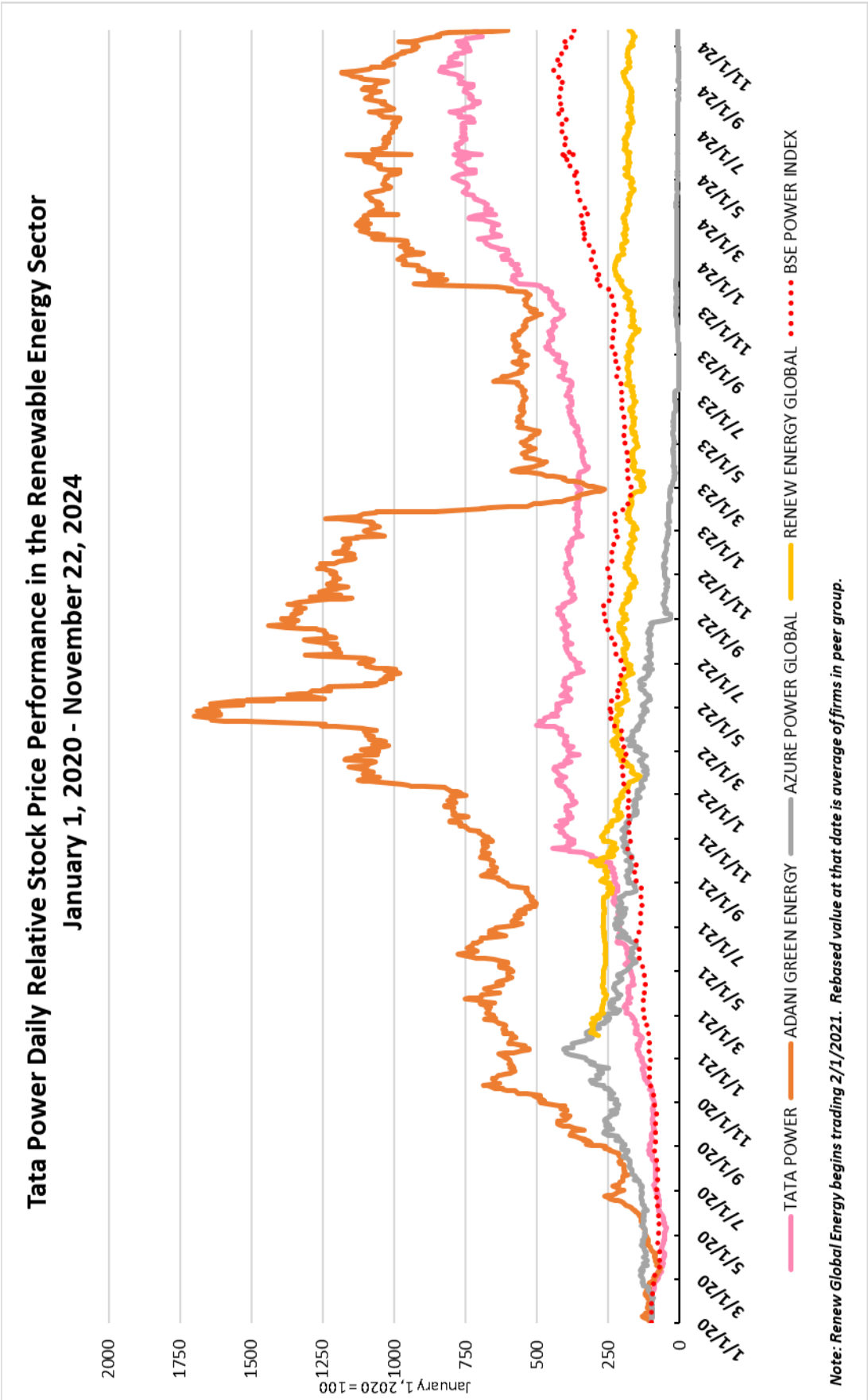
Source: Compiled by Aindrila Sinha from FY 2024 Annual reports of Tata Power, Adani Power, NTPC, and JSW Energy.

Exhibit 7B Stock Price Comparison of Tata Power With its Peers: Integrated Energy Companies (2016-20 and 2020-24) and Renewable Energy Companies (2020-24)



Tata Power Monthly Relative Stock Price Performance in the Integrated Energy Sector January 31, 2020-November 22, 2024





Source: LSEG Datastream, accessed 25th November, 2024

Note: The BSE Power Index is a stock market index that tracks the performance of leading companies in India's power sector.

Exhibit 8 Examples of Distressed Thermal Power Plants Acquired by Tata Power and Peers.

Company Name	Acquired Company/Distressed Thermal Plant Name	Acquisition Year	Capacity	Transaction Value (\$ million)
Adani Power	Coastal Energen	2024	1200 MW	402.22
JSW Energy	Ind Barath	2022	700 MW	126.59
Resurgent Power Ventures (a joint venture of Tata Power and ICICI Bank)	Prayagraj Power Generation Co	2019	1980 MW	724.73
Adani Power	GMR Chhattisgarh	2019	1370 MW	426.38
NTPC	Kanti Bijlee Utpadan Nigam & Nabinagar Power Generation Co.	2018	1110 MW	362.36
Adani Power	Korba West Power	2014	600 MW	507.31
Tata Power	Ideal Energy Project's Maharashtra Power Plant	2014	540 MW	362.36

Source: Compiled by Aindrila Sinha, Research Associate, HBS India Research Center, from JSW Energy. "JSW Energy Acquires Ind-Barath Energy (Utkal) Limited." <https://www.jsw.in/energy/jsw-energy-acquires-ind-barath-energy-utkal-limited>, accessed November 2024.

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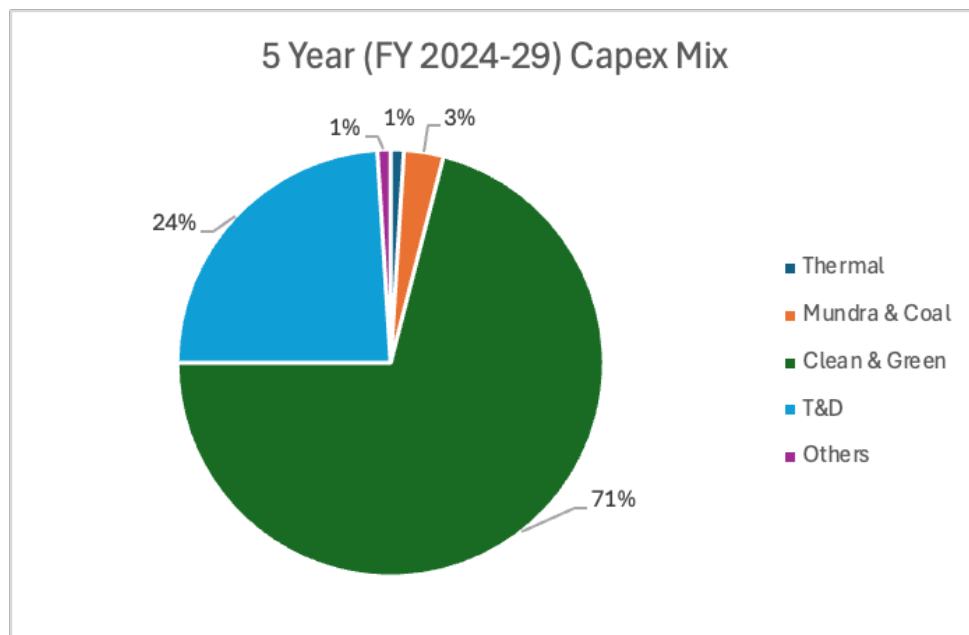
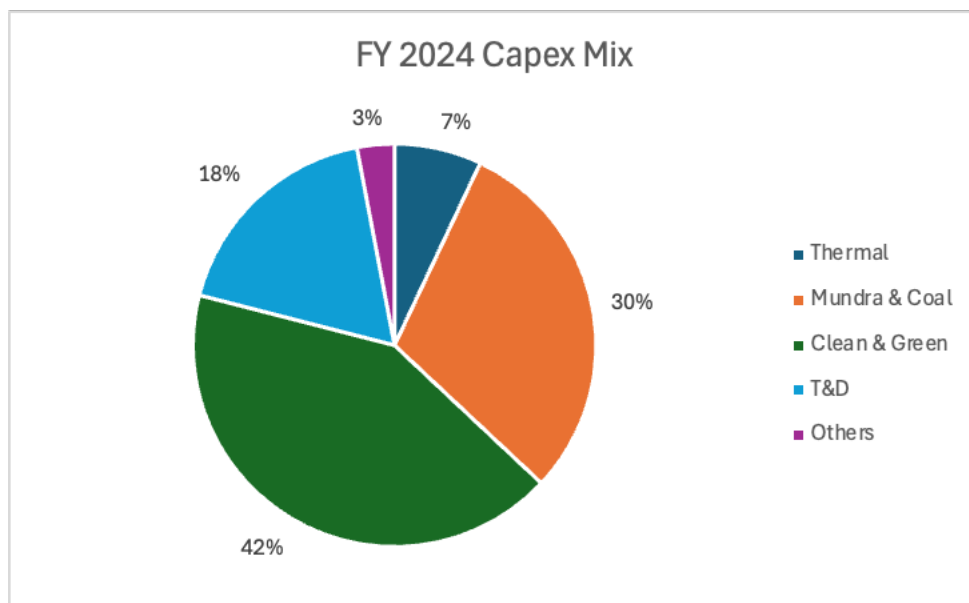
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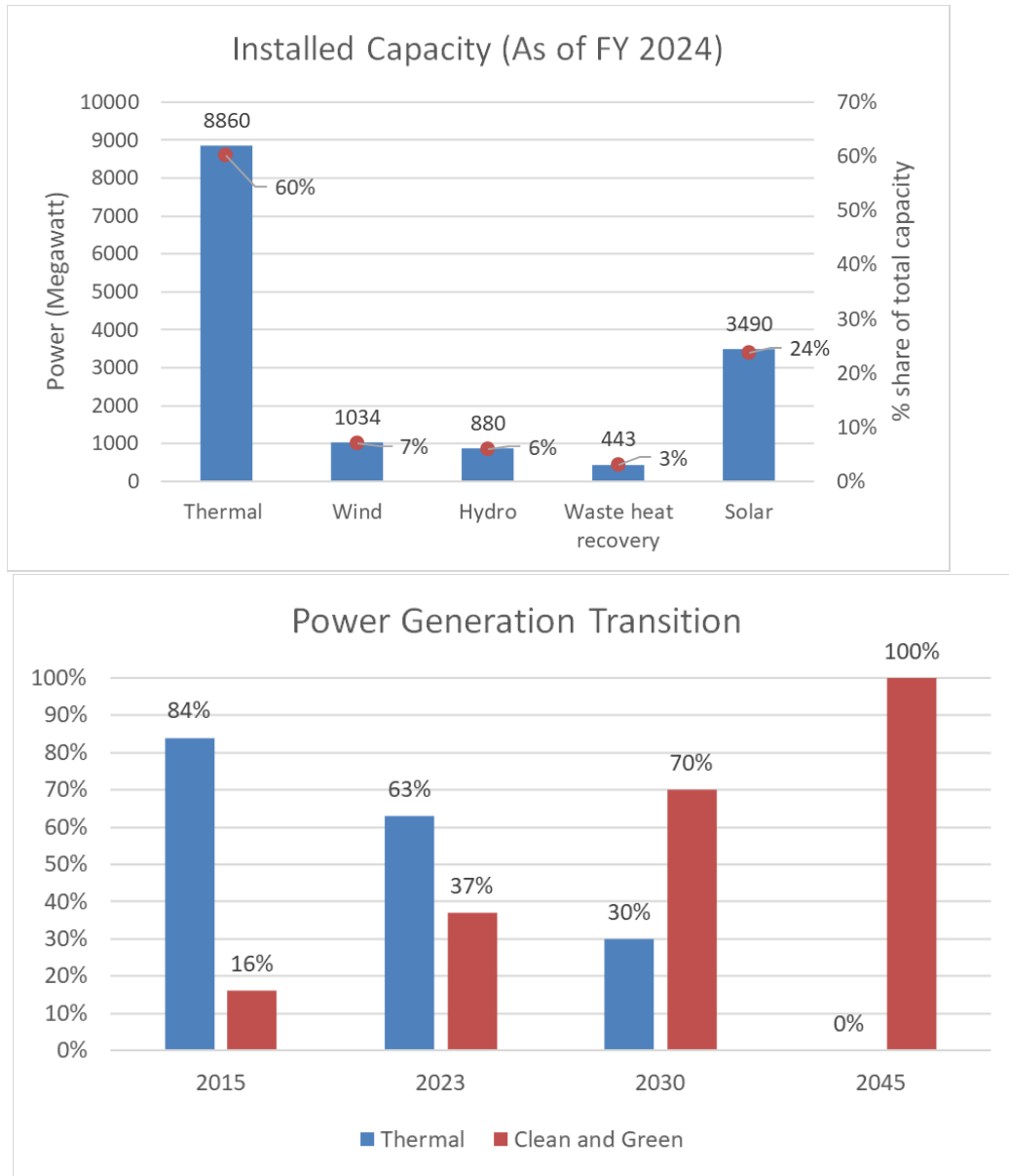
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Note: Conversion rate used: 1 USD=82.79 INR

Exhibit 9 Tata Power's Capex Breakdown in FY2024 and Five-Year Forecast (FY2024-2029)

Source: Tata Power Annual Report, <https://www.tatapower.com/content/dam/tatapoweraemsitesprogram/tatapower/pdf-root/company-financials/annual-reports/105AnnualReport-2023-24.pdf> (Pg 78)

Exhibit 10 Tata Power – Installed Capacity by Source and Tata Power Evolving Generation Mix

Source: Company Documents.

Exhibit 11 Examples of Reduction in AT&C Losses by Private Firms.

Company	State	Commencement Year	AT&C Losses (%)	
			Base Year (Y1)	Fifth year (Y5)
BSES Yamuna	Delhi	2001	60.6	45.6
BSES Rajdhani	Delhi	2001	49.6	37.8
Tata Power Delhi	Delhi	2001	46.4	26.5
Torrent Power Surat	Gujrat	2010	9.7	3.6
Torrent Power Ahmedabad	Gujrat	2010	14.3	6.3
Torrent Power Bhiwandi	Maharashtra	2007	58	19.3
Adani	Maharashtra	2018	11	6.7
CESC	Rajasthan	2016	29.7	20.4
CESC	West Bengal	1978	11.3	7.9
Torrent Power	Uttar Pradesh	2010	58.7	43.5

Source: Adapted by casewriters from Richard M. Rossow, Akshat Singh, "India's Struggling DISCOMs: Progress and Pitfalls," Center for Strategic and International Studies (CSIS), January 2024, p. 19-20, https://csis-website-prod.s3.amazonaws.com/s3fs-public/2024-01/240122_Rossow_India_Discoms.pdf?VersionId=rLQag_1y6cCYrggMbTN1jTcdQVVN2ia_, accessed October 2024.

Note: The base year (Y1) represents the baseline, showing the losses incurred when the utility was initially taken over by the private sector distribution company. For comparison, the fifth year (Y5) states the AT&C losses five years after the company assumed control.

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